

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

WAI4 - KGHS - KOREAN

Issue Date 05-Sep-2019

Revision Date 23-May-2023

Revision Number 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Name pH 7.00 Buffer

Product No 910107

Pure substance/mixture Mixture

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No information available

Supplier

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T 82.2.2023.0600

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Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

Made in USA

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SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Skin Corrosion/Irritation

Category 2

Serious Eye Damage/Eye Irritation

Category 2

Specific target organ toxicity - (repeated exposure)

Category 2

Environmental hazards

Based on available data, the classification criteria are not met

Chronic aquatic toxicity

Category 3

Label Elements



Signal Word

Warning

Hazard Statements

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H373 - May cause damage to organs through prolonged or repeated exposure

H412 - Harmful to aquatic life with long lasting effects

Precautionary Statements

Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P260 - Do not breathe dust/fume/gas/mist/vapours/spray

P273 - Avoid release to the environment

Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P321 - Specific treatment (see supplemental instructions on the administration of antidotes on this label)

P332 + P313 - If skin irritation occurs: Get medical advice/attention

P362 + P364 - Take off contaminated clothing and wash it before re-use

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

P314 - Get medical advice/attention if you feel unwell

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

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Other Hazards

This product does not contain any known or suspected endocrine disruptors

NFPA

Health

-

Flammability

-

Instability

-

Physical hazards

N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS No	Index No	Weight percent
Water	No information available	7732-18-5	KE-35400	95 - 97.5
Phosphoric acid, disodium salt	No information available	7558-79-4	KE-12344	0.5 - 1
Phosphoric acid, potassium salt (1:1)	No information available	7778-77-0	KE-28622	0.5 - 1
5-Bromo-5-nitro-1,3-dioxane	No information available	30007-47-7	KE-03687	0.5 - 1
C.I. Acid Yellow 23	No information available	1934-21-0	KE-06857	<0.1

SECTION 4: FIRST AID MEASURES

Description of first aid measures

General Advice

Use first aid treatment according to the nature of the injury. Get medical attention immediately if symptoms occur. Show this safety data sheet to the doctor in attendance.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur.

Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

Self-Protection of the First Aider

No special precautions required.

Most important symptoms and effects, both acute and delayed

Most important symptoms and effects

None reasonably foreseeable

Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically

SECTION 5: FIREFIGHTING MEASURES

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

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Unsuitable Extinguishing Media

No information available

Specific Hazards Arising from the Chemical

No information available.

Explosion Data

Sensitivity to Mechanical Impact None

Sensitivity to Static Discharge None

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required.

Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Vapours may accumulate to form explosive concentrations.

Methods and Material for Containment and Cleaning Up

Methods for Containment

Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labelled containers.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Handling

Wear personal protective equipment/face protection

Ensure adequate ventilation

Avoid contact with skin, eyes or clothing

Avoid ingestion and inhalation

Conditions for Safe Storage, Including any Incompatibilities

Storage

Keep container tightly closed in a dry and well-ventilated place

Store at room temperature in the original container

Protect from direct sunlight

Incompatible Products

No information available

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS No	Korea	ACGIH TLV	OSHA PEL
Water	7732-18-5	Not listed	Not listed	Not listed
Phosphoric acid, disodium salt	7558-79-4	Not listed	Not listed	Not listed
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not listed	Not listed	Not listed
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not listed	Not listed	Not listed
C.I. Acid Yellow 23	1934-21-0	Not listed	Not listed	Not listed

Component	CAS No	European Union	The United Kingdom	Germany
Water	7732-18-5	Not listed	Not listed	Not listed

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Phosphoric acid, disodium salt	7558-79-4	Not listed	Not listed	Not listed
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not listed	Not listed	Not listed
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not listed	Not listed	Not listed
C.I. Acid Yellow 23	1934-21-0	Not listed	Not listed	Not listed

ACGIH - Biological Exposure Indices

Component	CAS No	ACGIH - Biological Exposure Indices
Water	7732-18-5	Not listed
Phosphoric acid, disodium salt	7558-79-4	Not listed
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not listed
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not listed
C.I. Acid Yellow 23	1934-21-0	Not listed

Appropriate engineering controls

Engineering Measures None under normal use conditions

Individual protection measures, such as personal protective equipment

Personal protective equipment Use only those certified by the Korea Occupational Safety and Health Administration.

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur: Face protection shield.

Skin and Body Protection Wear protective gloves/protective clothing.

Respiratory Protection None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Light yellow
Odour	None
Odour Threshold	No information available
pH	7.00
PH Range	5.8 - 8.2

Property	Values	Remarks • Method
Melting point/freezing point	No information available	
Boiling point/range	~ 100 °C / 212 °F	
Flash Point	N/A	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit	No information available	
Vapour pressure	No information available	
Vapour Density	No information available	
Specific Gravity	No information available	

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Water Solubility	Soluble
Solubility in other solvents	No information available
Partition coefficient	No information available
Autoignition Temperature	-
Decomposition Temperature	No information available
Kinematic viscosity	No information available
Dynamic viscosity	No information available
Explosive Properties	No information available
Oxidising Properties	No information available

Other Information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

Reactivity

No information available

Chemical Stability

Stable under normal conditions

Possibility of Hazardous Reactions

None under normal processing

Conditions to Avoid

Extremes of temperature and direct sunlight.

Incompatible Materials

No information available

Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapours.

SECTION 11: TOXICOLOGICAL INFORMATION

Product Information

Information on likely routes of exposure

Inhalation	Not an expected route of exposure.
Ingestion	No known effect based on information supplied.
Eyes	Not an expected route of exposure. Irritating to eyes
Skin	No known effect based on information supplied.

Irritating to skin

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water 7732-18-5	LD50 > 90 mL/kg (Rat)	-	-
Phosphoric acid, disodium salt 7558-79-4	LD50 = 17 g/kg (Rat)	-	-
Phosphoric acid, potassium salt (1:1) 7778-77-0	LD50 = 3200 mg/kg (Rat)	-	LC50 > 0.83 mg/L (Rat) 4 h
5-Bromo-5-nitro-1,3-dioxane	LD50 = 455 mg/kg (Rat)	-	-

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30007-47-7			
C.I. Acid Yellow 23 1934-21-0	LD50 > 2000 mg/kg (Rat)	-	-

Information on Toxicological Effects

Symptoms No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Serious eye damage/eye irritation Irritating to eyes.
Sensitisation No information available

Mutagenic Effects No information available

Carcinogenicity No information available.

Reproductive Effects No information available

STOT - single exposure No information available

STOT - repeated exposure Causes damage to organs through prolonged or repeated exposure

Aspiration hazard No information available

Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral) 38924 mg/kg

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity

3.01% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Persistence and Degradability

Bioaccumulation

No information available

Mobility

Will likely be mobile in the environment due to its water solubility.

Component	log Pow
5-Bromo-5-nitro-1,3-dioxane 30007-47-7	1.6
C.I. Acid Yellow 23 1934-21-0	-1.572

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste disposal methods Disposal should be in accordance with applicable regional, national and local laws and regulations.

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Contaminated Packaging Improper disposal or reuse of this container may be dangerous and illegal.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport Not Regulated

IATA Not regulated

IMDG/IMO Not regulated

Marine Pollutant Not Applicable

Special Precautions for User No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	AICS
Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	X	X
Phosphoric acid, disodium salt	7558-79-4	KE-12344	X	231-448-7	X	X	-	X	X	X
Phosphoric acid, potassium salt (1:1)	7778-77-0	KE-28622	X	231-913-4	X	X	-	X	X	X
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	KE-03687	X	250-001-7	X	X	-	X	-	X
C.I. Acid Yellow 23	1934-21-0	KE-06857	X	217-699-5	X	X	-	X	X	X

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Water	7732-18-5	Listed	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Listed	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Listed	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable

Korean National Regulations

Component	CAS No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Annex 1 - KE-12344	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Annex 1 - KE-28622	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Annex 1 - KE-03687	Not applicable	Not applicable

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C.I. Acid Yellow 23	1934-21-0	Annex 1 - KE-06857	Not applicable	Not applicable
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Component	CAS No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable

Component	CAS No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable

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National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS No	Korea	ACGIH - Biological Exposure Indices
Water	7732-18-5	Not listed	Not listed
Phosphoric acid, disodium salt	7558-79-4	Not listed	Not listed
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not listed	Not listed
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not listed	Not listed
C.I. Acid Yellow 23	1934-21-0	Not listed	Not listed

US Management Information

OSHA Occupational Safety and Health Administration

Not applicable

Component	CAS No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Water	7732-18-5	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Component	CAS No	CERCLA EHS RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt	7558-79-4	Not applicable	5000 lb	Not applicable
Phosphoric acid, potassium salt (1:1)	7778-77-0	Not applicable	Not applicable	Not applicable
5-Bromo-5-nitro-1,3-dioxane	30007-47-7	Not applicable	Not applicable	Not applicable
C.I. Acid Yellow 23	1934-21-0	Not applicable	Not applicable	Not applicable

GHS Classification

Based on available data, the classification criteria are not met.

EUH210 - Safety data sheet available on request.

SECTION 16: OTHER INFORMATION

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Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.©
Issue Date	05-Sep-2019
Revision Date	23-May-2023
Reason for revision	Initial Release.

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Disclaimer

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet